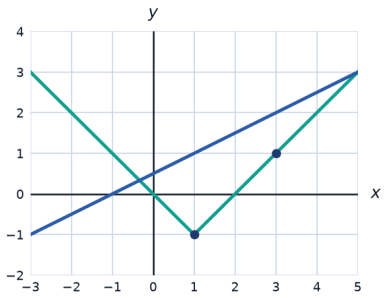
# Optimization and Curve Sketching

## Section 1: Critical Points and First Derivative Test

**1** Find all critical points of $f(x) = 2x^3 - 9x^2 + 12x - 5$.




**a.** $x = 1$ and $x = 2$

**b.** $x = -1$ and $x = -2$

**c.** $x = 3$ and $x = 0$

**d.** $x = 2$ only

**2** Classify the local extrema of $f(x) = 2x^3 - 9x^2 + 12x - 5$.

**a.** Local max at $x = 1$; local min at $x = 2$

**b.** Local min at $x = 1$; local max at $x = 2$

**c.** Inflection point at $x = 1$

**d.** Local max at $x = 2$ only

**3** Find the absolute maximum of $g(x) = 4x - x^2$ on the closed interval $[0, 3]$.

**a.** $4$

**b.** $3$

**c.** $0$

**d.** $5$

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## Section 2: Concavity and the Second Derivative Test

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**4** Determine the interval(s) where $f(x) = x^4 - 4x^3 + 10$ is concave upward.

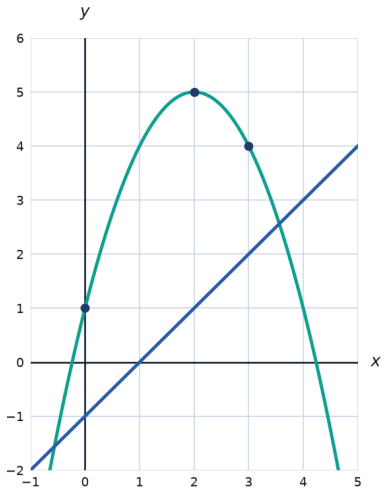
**a.** $(-\infty, 0) \cup (2, \infty)$

**b.** $(0, 2)$

**c.** $(2, \infty)$

**d.** $(-\infty, 2)$

**5** Find the $x$-coordinates of all inflection points of $y = x^3 - 6x^2 + 9x$.




**a.** $x = 2$

**b.** $x = 1$ and $x = 3$

**c.** $x = 0$

**d.** $x = 4$

**6** If $f'(3) = 0$ and $f''(3) = -5$, what does the Second Derivative Test conclude about $x = 3$?

**a.** $f$ has a local maximum at $x = 3$

**b.** $f$ has a local minimum at $x = 3$

**c.** $x = 3$ is an inflection point

**d.** The test is inconclusive

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## Section 3: Applied Optimization: Perimeter, Area, and Volume

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**7** A farmer has $120$ m of fencing to enclose a rectangular field adjacent to a river (no fence needed along the river). What dimensions maximize the enclosed area?

**a.** Width $= 30$ m, Length $= 60$ m

**b.** Width $= 40$ m, Length $= 40$ m

**c.** Width $= 20$ m, Length $= 80$ m

**d.** Width $= 25$ m, Length $= 70$ m

**8** An open-top rectangular box is made from a $16$ by $10$ inch cardboard sheet by cutting equal squares of side $x$ from each corner and folding up the sides. What $x$ maximizes the box volume?

**a.** 2 inches

**b.** 3 inches

**c.** 1.5 inches

**d.** 2.5 inches

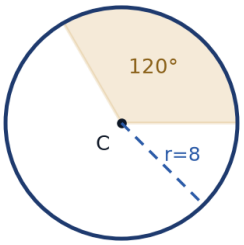
**9** A closed cylindrical can must hold $1000$ cm$^3$ of liquid. What relationship between height $h$ and radius $r$ minimizes total surface area?

**a.** $h = 2r$

**b.** $h = r$

**c.** $h = 4r$

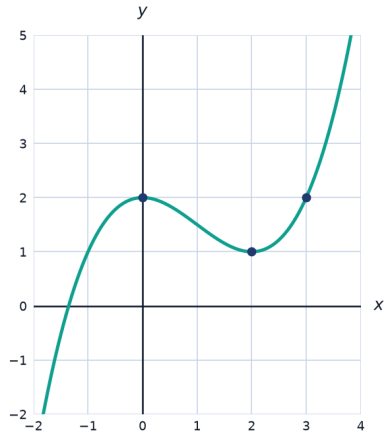
**d.** $h = \pi r$




## Section 4: Economics and Distance Optimization

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**10** A company's revenue and cost functions are $R(x) = 50x - 0.5x^2$ and $C(x) = 10x + 200$. What production level $x$ maximizes profit?




**a.** 40

**b.** 50

**c.** 30

**d.** 60

**11** Find the point on the line $y = 2x + 1$ that is closest to the point $(4, 4)$.

**a.** $(2, 5)$

**b.** $(1, 3)$

**c.** $(3, 7)$

**d.** $(1.5, 4)$

**12** A Norman window has the shape of a rectangle surmounted by a semicircle. If the perimeter is $16$ ft, what radius $r$ of the semicircle maximizes the light admitted?

**a.** $\frac{16}{4+\pi}$ ft

**b.** $\frac{8}{2+\pi}$ ft

**c.** 2 ft

**d.** $\frac{16}{\pi}$ ft

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